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Methodology for Assessing Existing Vulnerabilities in Modbus Protocol

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Abstract

Modbus protocol is one of the most extensively used communication protocols in Industrial Control Systems (ICS) because of its simplicity reliability, and basic position in ICS communication and availability as an open-source solution, However, the rising use of Ethernet Modbus TCP/IP connections and their integration with the internet have exposed this protocol to a variety of security concerns. A comprehensive assessment of related literature shows that the lack of encryption is a fundamental flaw in the Modbus protocol used in ICS systems. In this study, we evaluate the vulnerability of Modbus communication configuration by providing the methodology for assessing the existing vulnerability: Lack of Encryption, which includes an Allen-Bradley (1769-L30ER) Programmable Logic Controller (PLC) as the server and a Modbus Poll client (PC). The results revealed clearly that the sent data remained in plain text, making it vulnerable to interception and modification. In addition, we performed a Denial-of-Service (DoS) attack on the experimental setup to demonstrate how easily inside attackers may exploit this existing vulnerability. As a result, improving the security of the Modbus protocol is essential. This study attempts to throw attention on the critical need to improve Modbus protocol security mechanisms in order to protect ICS systems from possible attacks.

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Keywords: Modbus; Security; Industrial Control Systems; SCADA; Vulnerability Assessment;

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1. Introduction

Critical infrastructures, such as water treatment facilities, electricity grids, and transportation networks, are adopting increasingly modern technology, such as computers and sensors, to monitor and regulate physical processes. These technologies are incorporated into Industrial Control Systems (ICSs), which are networks used to monitor and control operations in real time. ICSs systems, in essence, offer remote monitoring and control of physical processes, which can improve efficiency, productivity, and safety in critical infrastructure industries(1). ICS (Industrial Control Systems) refers to a range of control systems used in industries. SCADA, DCS, PLC, and HMI are examples of industrial control systems that fall under the umbrella of ICS. These systems are often encountered in the process control sectors of the chemical and petrochemical industries. In 2013, the German government unveiled the "Industry 4.0" plan, which intends to transform conventional sectors using artificial intelligence. The purpose is to enable intelligent remote control and operation of industrial facilities, resulting in higher production. Industrial operations are monitored and controlled using SCADA (Supervisory Control and Data Acquisition) systems. DCS (Distributed Control Systems) are systems that are like SCADA but are utilized in bigger industrial processes. PLCs (Programmable Logic Controllers) are used to automate industrial processes, whereas HMIs (Human Machine Interfaces) are graphical user interfaces that allow operators to interact with the control system. The industry 4.0 approach seeks to increase the efficiency and productivity of industrial operations via the application of artificial intelligence. This entails the use of sophisticated remote control and operation of industrial facilities(2). Communication takes place in an ICS between various components and subsystems in order to share information and regulate physical processes. Sensors, actuators, controllers, HMIs, SCADA, input/output components, and communication networks are examples of these components. All of these components are linked together by a network, allowing them to interact and share data in real time(3). The processors are the ICS system's brains, controlling the system's operational elements. They process data from numerous components and make decisions based on that data to ensure the system functions smoothly and safely. The system's input/output components allow it to interface with external devices such as sensors and actuators. These components collect data on physical processes and send it to processors for analysis. These data are then used by the processors to make judgements and transmit orders to the actuators, which control the physical processes. The graphical interface that allows the operator to interact with the system is known as the HMI. It displays a visual depiction of the system and allows the operator to monitor its performance and modify its parameters. SCADA software collects data from input/output components and offers vital monitoring and control of industrial processes. It enables operators to monitor and manage the system remotely, ensuring that vital infrastructure systems operate effectively and safely. Finally, communication networks connect the various ICS system components, allowing them to interact with one another and with external devices. This seamless component integration guarantees that the ICS system can efficiently operate and monitor critical infrastructure systems. (4) Industrial control systems (ICS) are frequently linked to a corporate network used for day-to-day company operations through Ethernet. The control system network, however, is isolated from the corporate network via a security gateway to protect its security. This isolation is achieved by employing various permission mechanisms, firewalls, and virtual LANs (5).The communication can take place through wired or wireless connections, and the protocols utilized are determined by the ICS in use. To allow communication across multiple devices, ICS often employ established communication protocols. These protocols let devices to communicate data and coordinate their operations, which is critical for ICS systems to function well. Among the most prevalent communication protocols used in ICS are: MODBUS, DNP3, ETHERNET IP, IEC 60870-5-104, IEC 61850 , HART , S7 COMPROMISED, PROFIBUS, BACnet and OPC-UA (6) (7). The Modbus protocol is extensively used in ICS systems and was initially developed for serial communication, with the assumption that all ICS operations are secure and functioning properly. However, because there are no authentication or permission measures built into the protocol, and data is sent in plaintext with no encryption, Modbus systems are vulnerable to criminal activities(8). According to HMS Networks (Industrial network market shares) in 2021 (9) the market share of various Industrial Ethernet protocols will be calculated. According to the survey, Ethernet/IP and Profinet are the most prevalent protocols, accounting for 17% of the market, followed by Ether Cat (7%). Fieldbus protocols such as Profibus and Device Net, on the other hand, have lost 5% of their market share in comparison to the previous year. While being actively used in testing, the Modbus protocol (TCP and RTU variations) only has a 10% market share (5% RTU, 5% TCP). Nonetheless, it is still the most often used protocol in testbeds, datasets, and Internet measurements. The Modbus transmission protocol lacks security

features and has become insecure as the number of closed ICSs linked to the internet has increased, making them an ideal target for cyber attackers(10).

Modbus protocol, a form of message protocol used for device communication. It works at the application layer; therefore, it is unaffected by the underlying network protocol. One of the reasons for its success in industry is because it is adaptable to both serial and routable protocols. Because the protocol is straightforward and does not rely on authentication, there is very little overhead while interacting. As a result, it is well suited to compact devices with low computing power. Modbus is a device communication protocol that is designed to interact with a variety of networks. It is simple to use and requires little computing power, making it a popular ICS systems(7). Modbus is a message exchange protocol for the application layer that has been widely used in the industry since 1979 by Gould Modicon (now Schneider). It allows communication between millions of automation devices and may be accessible by the internet community via the TCP/IP stack's reserved system port 502. Modbus is a request/response protocol that provides services that are described by function codes. It enables simple communication across all network topologies, and any device, including PLCs, HMIs, Control Panels, Drivers, Motion Control, I/O Devices, and so on, may utilize the Modbus protocol to commence a remote activity(11).The client-server or master-slave communication model is used in this protocol, in which one device, the master station, starts communication with another device, the slave station. The master station issues a request to the slave station, which responds to the master station. This protocol allows for one-to-one as well as many-to-many communication (12).For implementing the Modbus protocol, numerous transmission methods are available, including asynchronous serial transmission(Modbus RTU, Modbus ASCII,), TCP/IP over Ethernet, and Modbus Plus(13).

Modbus/TCP is a communication protocol used to link devices such as sensors, actuators, and controllers in industrial automation systems. An Application Data Unit (ADU) is a data structure defined by the Modbus/TCP protocol that is used to communicate information between devices (12).The Modbus TCP Application Data Unit (ADU) is made up of two parts: a 7-byte Modbus Application Protocol Header (MBAP) and an N-byte Protocol Data Unit (PDU). The MBAP includes data such as the source and destination addresses, the length of the PDU, and the transaction identification. The data being communicated between the devices is stored in the PDU. Devices may communicate data simply and effectively utilizing this protocol, allowing automation systems to run smoothly(14) The inclusion of the MBAP header distinguishes a typical Modbus PDU from a Modbus/TCP ADU. The MBAP header adds information about the data being transported that is required for communication over a TCP/IP network. Devices in Modbus are recognized by unique IDs, and errors are checked using a Cyclic Redundancy Check (CRC). Modbus/TCP, on the other hand, employs a Modbus Application Protocol (MBAP) header that contains numerous elements such as Transaction Identifier, Protocol Identifier, Length, and Unit Identifier. In Modbus/TCP, TCP/IP is utilized for error checking. The Modbus version used is determined by specific application requirements such as the kind of communication channel, distance between devices, and needed data transmission rates(15). The Modbus data packet frame shown in Fig. 1.

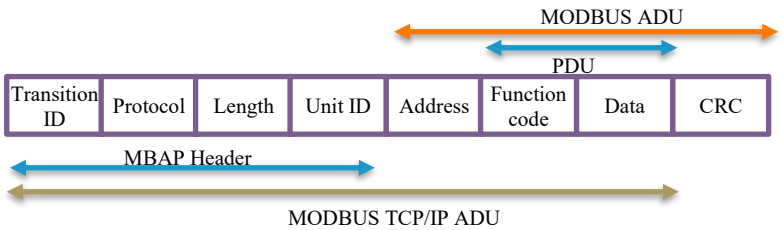

Fig. 1. Modbus Data Packet Frame

2. Security issues in the Modbus protocol

The Modbus protocol, which is used in ICS networks, lacks fundamental security features. As a result, it is vulnerable to several sorts of attacks, including man-in-the-middle attacks, eavesdropping attacks, and replay attacks, Denial of services (DOSs), that might comprise the system's confidentiality, integrity, and availability. These flaws can have a substantial impact on businesses and the general public, particularly when vital infrastructure assets such

as power plants, water distribution networks, and train transportation systems are attacked(11). This implies that the Modbus protocol used to communicate across different devices in the ICS lack a mechanism for verifying the identity of the transmitter or recipient of data. As a result, attackers may simply tamper with, steal, or replay data without detection. The data sent through this protocol is in plaintext, which means it is not encrypted and may be readily intercepted and read by anybody with access to the communication channel. This makes it easier for attackers to exploit protocol vulnerabilities and conduct out attacks on ICS(16) . The Modbus protocol, used in industrial control systems, is not immune to security flaws. Attackers can take advantage of these flaws to obtain unauthorized access to the system or modify its functions. The following are some of the most popular Modbus protocol attacks shown in Table 1 with their description.

Table 1. Attacks with Their Description Exploiting the vulnerabilities in Modbus protocol.

Attack	Description	Identification	Prevention
Man-in-Middle Attack (MITM)(17) (18)	The MitM attack intercepts Modbus data among two devices in order to obtain details about their use and design and possibly launch additional assaults.	Ettercap tool for OpenPLC ARP cache spoofing.	strong authentication mechanisms, intrusion detection (IDS) and prevention systems, and regular system updates
Denial of services (DoS)(18)	This includes overloading Modbus devices with traffic, as delivering many erroneous or invalid requests or flooding the network with a high volume of data. Causing system breakdowns and interrupting industrial operations,	The Ettercap interface was used to detect the Attack.	Using firewalls IDS, communication methods like Transport Layer Security (TLS)
Eavesdropping Attack(19)	Attack on Modbus is detecting and collecting Modbus data among two devices in order to obtain private information, accomplished by analyzing the contents of Modbus packets with a packet sniffer or network monitoring tool.	-	-
Message Spoofing Attack(20)	A Modbus message spoofing attack includes forging or manipulating packets in order to convey misleading or malicious commands. This can be accomplished by leveraging Modbus communication flaws such as poor authentication systems or a lack of data integrity checking.	HMAC	stream control transmission protocol (TSP) and hash-based message authentication code (HMAC) technologies are used.
Reconnaissance attack(21) (18)(22)	Modbus attacks involve gathering information about Modbus devices and are carried out using various methods such as port scanning, network mapping, and banner grabbing to detect Modbus devices on a network and extract information about their operating system, Modbus slave ID, or supported function codes.	intrusion detection approach based on automated learning for detection is used.	-
Replay attack (23)	This attack entails intercepting and re-transmitting the Modbus protocol packets by recording Modbus traffic and replaying it later, or by altering recorded packets to alter their content in order to execute unauthorized commands or transactions, potentially causing operational disruptions or damage.	By monitoring the sequence numbers and timestamps in Modbus connections.	-
Response Injection attack (23) (18)	Intercepting Modbus answers to include falsified data or commands, such as modifying register values, beginning unauthorized transactions, or forcing the device to reset, and inserting malicious data or commands into them to perform unauthorized Modbus device activities. An attacker normally requires access to the Modbus communication channel, which can be obtained by eavesdropping or acquiring access to a compromised device.	-	frame filtering module is used preventing unwanted attacks such as False Command Injection and False Access Injection from accessing the programmable logic controller (PLC).
Network Scanning(10)	Attacker can use tools like Nmap or Metasploit to search a network for devices that implement the Modbus protocol, and then analyze the answers to Modbus-specific queries.	IDS	Intrusion Prevention system (IPS)

Sniffing (24)	Attacker can intercept Modbus communication on a network by using a packet sniffer such as Wireshark. The attacker can identify if a device is using Modbus by analyzing what's inside of the packets.	Snort and Syslog-ng programmes are used to detect the sniffing attack.	-
Honeypot (10)	An attacker can place a fake Modbus device (honeypot) on a network to attract other Modbus devices and then analyze the traffic to learn more about those devices.	-	A modular security test system is established to examine the security of a system.

3. Related work

The Modbus protocol has received substantial attention in the realm of industrial control systems (ICS), particularly in terms of security flaws. Because of its broad use and possible hazards, several academics have recognized the need of protecting Modbus communication. In this part, we give a detailed assessment of the available literature on Modbus protocol security. Our literature research covers a wide variety of issues, including but not limited to vulnerability identification, assessment of attack vectors, defence mechanism analysis, and evaluation of security solutions suggested for Modbus communication.

May Bashendy et.al in paper (25) proposes a complete attack tree model for the Modbus/TCP protocol, which covers several sorts of attacks such as reconnaissance, man-in-the-middle, denial of service, and replay. Ayesha Rahman et.al finds Modbus protocol vulnerabilities and efficiently performs two DoS attack scenarios on a simulated programmable logic controller in research work of (17) .

The study in paper (18) focuses on the Modbus TCP/IP Protocol vulnerabilities found in Industrial Control Systems (ICS) and how these vulnerabilities might be exploited through particular attacks.

The research by El-Hawary et.al describes attacks against the recently announced authenticated Modbus protocol, in paper (26) which is utilized in power plant distributed control systems.

Rajesh et.al in paper(23) provided a method for detecting and stopping unwanted attacks on SCADA systems using the Modbus protocol. In same work (23)On a real-time SCADA testbed, the approach was found to stop 97% of malicious Modbus events or assaults from reaching the PLC. In future further Modbus-related hacks and offer techniques to improve the security of SCADA systems against those attacks.

Thomas H .et.al in paper (27) details 28 cyber assaults that may be launched against MODBUS-enabled industrial control systems. These assaults can cause significant damage to systems, resulting in financial loss and safety concerns. Research paper (28) provides a technique for detecting DoS or flooding attacks in Industrial Control Systems (ICS) using the MODBUS communication standard. The suggested technique detects user application-level flooding or denial-of-service (DoS) assaults and raises an alarm in the Supervisory Control and Data Acquisition system (SCADA) to alert administrators or engineers to take corrective action.

Tongxin Li et.al in research paper (29)describes the experimental findings of detecting DoS attacks in OpenPLC using a digital twin system with a Raspberry Pi and Snort Intrusion detection and the results demonstrate that the constructed digital twin system can identify DoS attacks in OpenPLC effectively.

Bo Chen et.al describes a real-time cyber-physical framework or test bed for investigating the vulnerabilities of power system communication protocols in paper (30) To investigate cyber and physical system vulnerabilities in smart power grids, the framework incorporates a real-time power system simulator and a communication system simulator

The authors of the paper (11) presents a new secure version of a role-based access control paradigm (RBAC). A security and performance analysis are performed on the suggested model that demonstrates that the concept is resistant to various types of attacks.

By employing upon the current knowledge, we contribute to the field of Modbus communication security in ICS systems. A detailed analysis of current literature on Modbus-based ICS systems is undertaken, as well as an evaluation of known risks. Key research problems are indicated based on the findings. Following that, a proposed framework for vulnerability assessment is given, with particular focus on the lack of encryption. The study contains an examination of experimental outcomes produced from the proposed structure, highlighting the need of strong and up-to-date security procedures in ICS systems.

4. Methodology adopted for vulnerability assessment in Modbus Protocol.

In our work on the Modbus protocol, we concentrated on implementation and security analysis. Several vulnerabilities linked with the Modbus protocol were uncovered throughout our research. To address these weaknesses, we examined several articles and research to obtain an in-depth understanding of the present challenges and potential threats. The lack of encryption in the Modbus protocol was identified as a key weakness throughout my examination. As a result, sensitive data transferred across the network may be intercepted and read by unauthorized persons, posing a possible security risk. The lack of encryption makes it easy for attackers to manipulate or eavesdrop on Modbus device traffic, potentially resulting in unauthorized control or data loss.

Fig. 2. Multiple Phases for Vulnerability Test

4.1. Lack of Encryption

We created a methodology flow Diagram that includes multiple phases to test one of the vulnerabilities i.e., lack of encryption in the Modbus protocol. The phases are shown in Fig. 2. Steps for Checking Existing Vulnerabilities i.e., lack of Encryption with Modbus Protocol Shown in Fig. 3.

4.2. Establish Client-Server Communication

To test the Modbus protocol's security, we created a client-server communication environment by establishing a Modbus poll as the communication client and a Programmable Logic Controller (PLC) named Allen-Bradley(1769-L30ER) as the server. This setup enables us to create connection between the client and server, replicating a real-world scenario interaction, and assessing the Modbus protocol's current vulnerability.

4.3. Program the Devices

We programmed the devices using Studio 500 through the Modbus protocol to simplify communication between the client and server. The client device (Modbus poll) delivers data reading and writing requests to the server PLC. The server device, is conFig.d to respond to these requests and conduct the necessary activities.

4.4. Analyze Traffic Using a Sniffing Tool

We use a sniffer tool (Wireshark) on the system executing Modbus poll to investigate the data transmitted between the client and server. The sniffer programme allows us to observe and record data transfer between the client and server on the chosen network. We acquire insights into the underlying data flow and potential vulnerabilities by monitoring the conversation between the client and server.

4.5. Monitor Communication (Read/write) data

We examine the Captured packets to determine the content, structure, and trends of Modbus protocol communication. Wireshark tool allows us to study the communication patterns and discover any security flaws or vulnerabilities. With the sniffer gear in place, we closely watch the data traffic on the Modbus protocol.

4.6. Identify Lack of Encryption

During the monitoring procedure, we performed a real-time analysis of the Modbus protocol communication data. This research revealed that there was a noticeable absence of encryption in the communication. We saw that the information being transferred in plain text when we monitored the data packets being exchanged between the client and server. This indicates that the data was not encrypted or otherwise safeguarded, leaving it vulnerable to unauthorized access and interception.

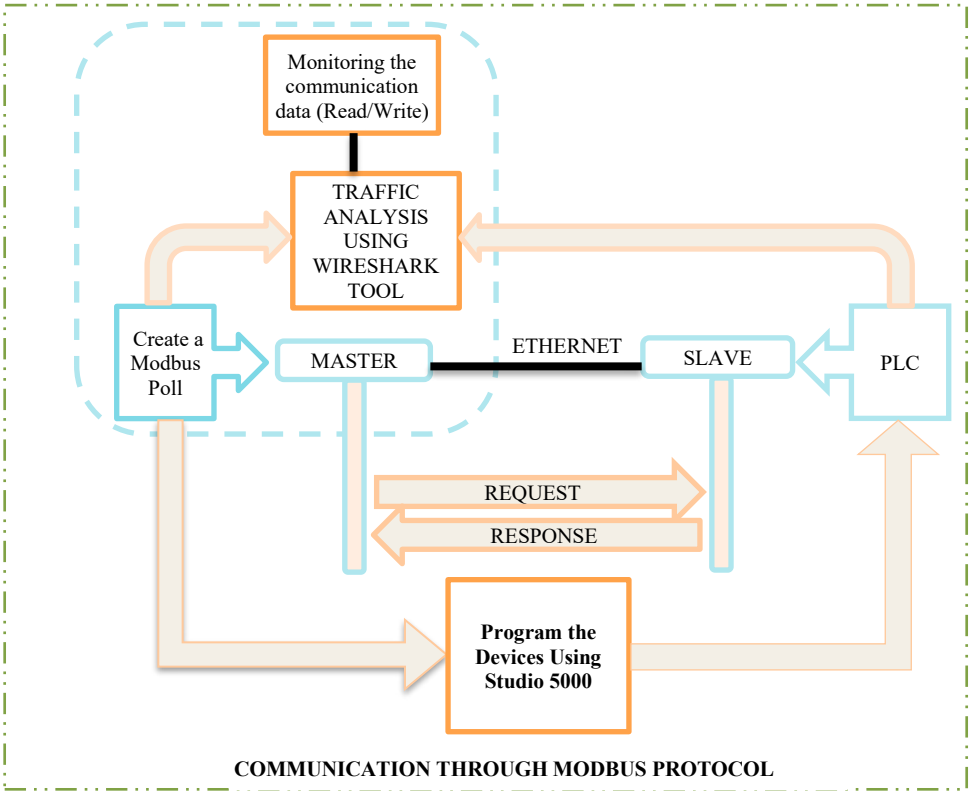

Fig. 3. Framework for Vulnerability Assessment of Modbus Protocol

4.7. DOS Attack

The method used for carrying out a Denial of Service (DoS) Attack at the CUJ SCADA lab by an inside attacker utilizing the hping3 tool is shown in Fig. 4. We use the Wireshark to Analyze the data transmission between the server (PLC-Allen-Bradley 1769-L30ER) and the client PC (Modbus Poll), which both have been connected to the same network and use the Modbus communication protocol. The attacker sends malicious data packets to the PLC server - Allen Bradley through Modbus Poll on PC in the Scada lab.

5. Results and Discussion

We successfully evaluated the vulnerability in the Modbus protocol using this technique by establishing client-server communication, configuring the devices, analyzing the traffic, monitoring data transfer, and discovering the lack of encryption. This comprehension emphasizes the significance of employing encryption measures to safeguard the Modbus connection and reduce potential security threats. We used a variety of tools and techniques to gain specific insights in our comprehensive vulnerability assessment targeting the absence of encryption in the Modbus protocol. To begin, we used ladder programming on a PC, which we installed as a Modbus client. This configuration allowed us to conduct numerous operations on the Programmable Logic Controller (PLC)Allen Bredily(1769-L30ER), as shown in Fig. 5.

As illustrated in Fig. 6. we used the client program Modbus Poll to create a secure Ethernet connection between the PC and the PLC Allen Bredily(1769-L30ER). We were able to precisely simulate data transfer scenarios and closely monitor the entire process due to this well-designed architecture. Fig. 7. shows the Modbus Poll software's configuration or setup, which is a tool for evaluating Modbus connectivity. The Modbus Poll and PLC connection shows that Modbus Poll is being utilized to interface with the PLC. The PLC address acts as a unique identification,

allowing Modbus Poll to connect to and interact with the PLC. This stage is part of the vulnerability evaluation process, and it examines the Modbus Poll connection and setup.

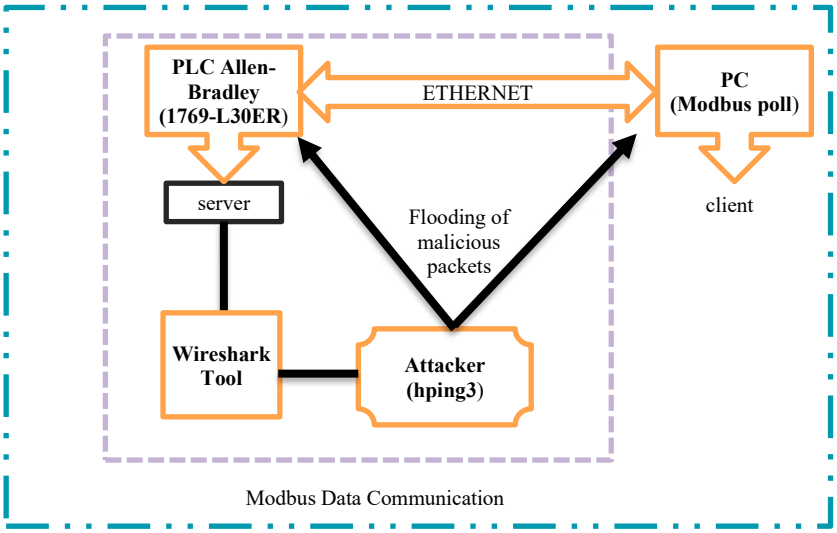

Fig. 4. Methodology for DoS attack in Modbus-based ICS

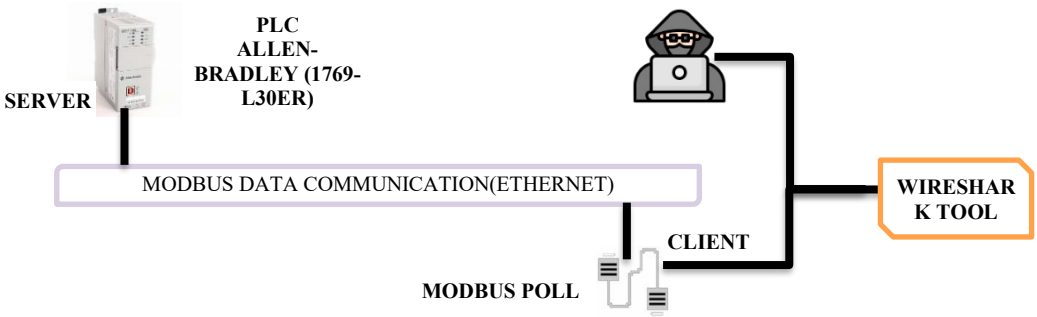

Fig. 5. Setup for the Vulnerability Assessment of Modbus Protocol

Wireshark, a network protocol analyzer, is used to examine the existing vulnerability. We may investigate the content and security of the connection by capturing and analyzing the data flow between the Modbus master (Modbus poll) and slave (PLC). Fig. 8. depicts the Wireshark interface, which displays data transmitted between the master and slave. A detailed examination of the data transfer in the Modbus protocol. We used Wireshark's statistics function to explicitly assess the presence or absence of encryption. We were able to determine the information being exchanged, as well as the Data registers utilized during the communication process, by carefully analyzing the information that was collected with the help of Wireshark revealed that no encryption was used. Following an in-depth examination of the Modbus protocol, it was discovered that there is a serious vulnerability in the form of a lack of encryption in data transfer.

This vulnerability poses a significant danger since an attacker can use it to obtain access to critical information about ICS (Industrial Control Systems) systems. An attacker can readily extract crucial data by doing traffic analysis on data packets. Fig. 9. depicts the information included within the data packets, including information on the registers being used. This discovery revealed a severe vulnerability, since malevolent attackers might intercept, modify, or exploit the sent data. These insights revealed the system's possible hazards and weaknesses, emphasizing the importance of encryption to protect data confidentiality and integrity.

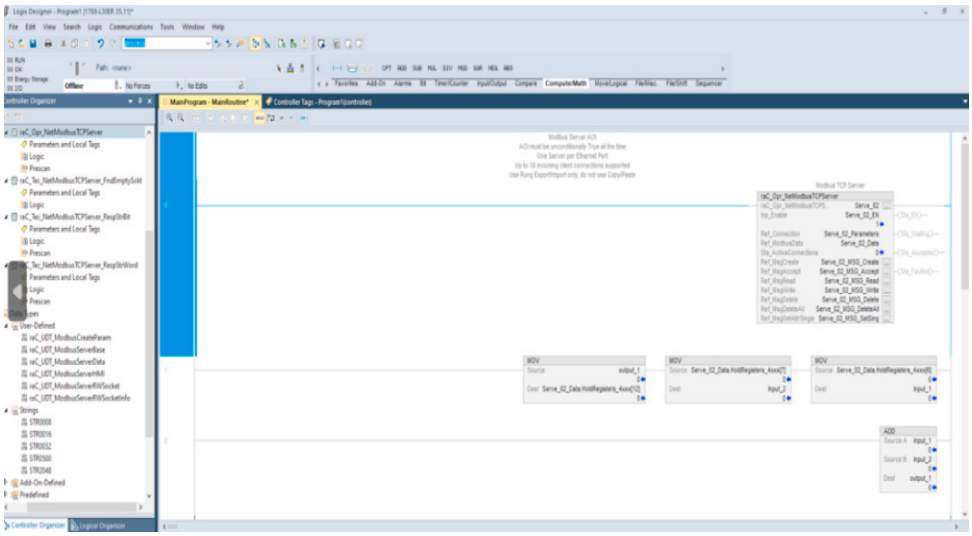

Fig. 6. PLC Programming in Studio 5000

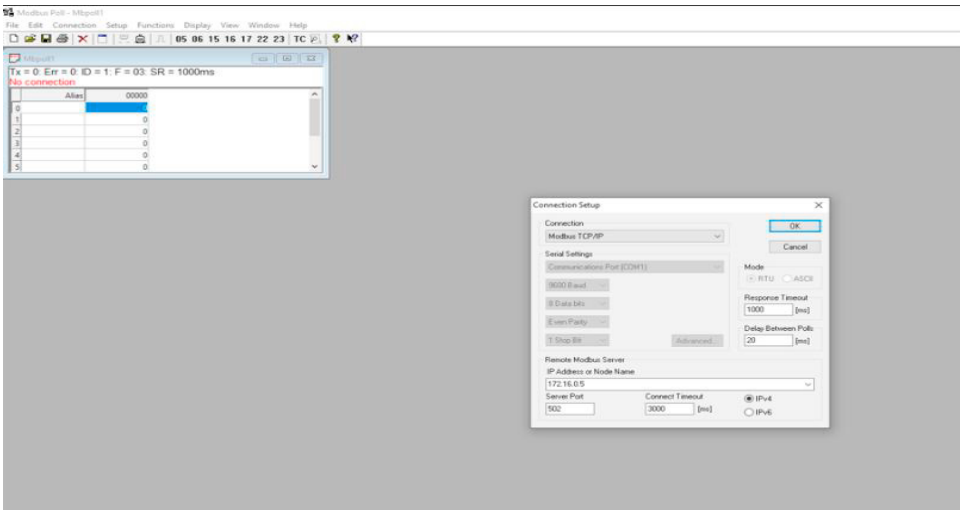

Fig. 7. Establishment of Connection between Modbus Poll and PLC

The lack of encryption allows an internal attacker to easily collect and analyze sent data, jeopardizing the security and integrity of ICS systems, with network access may easily exploit this vulnerability, possibly causing substantial damage.

Flood Attack Preparation: We identify vulnerabilities in the Modbus protocol or server implementation that can be used to launch a DoS attack. The hping3 tool is selected to carry out the flood attack. hping3 is a multi-purpose command-line program that may be used for network scanning, fingerprinting, and DoS attacks. Flooding Attack: The flood attack is launched by the attacker by using the hping3 command on PC. Hping3 begins sending many malicious messages to the PLC server over Modbus Poll. The flood attack reduces the server's resources, rendering it ineffective or extremely slow.

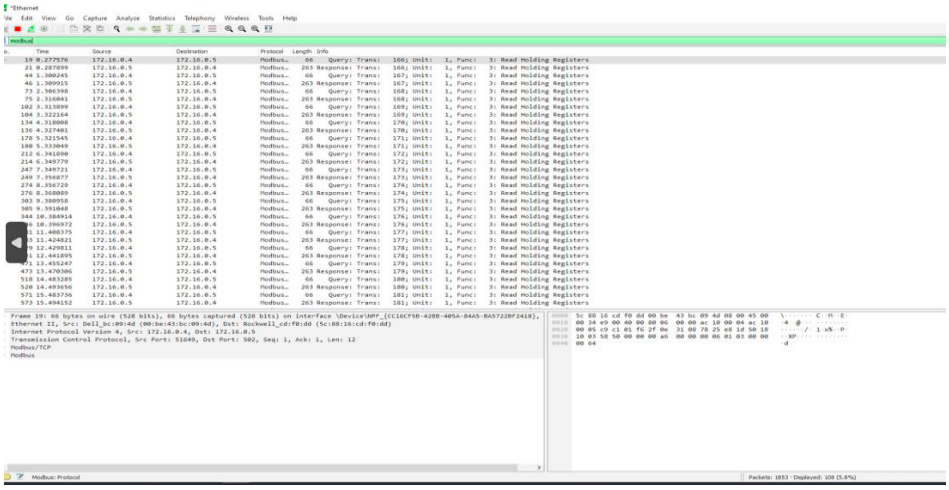

Fig. 8. Modbus protocol Packets captured with the help of Wireshark tool

Even the sever PLC (Allen-Bradley) are reduced to communication. The server PLC (Allen-Bradley) suffers from a Denial-of-Service problem as a result of the flood attack, making it unable to respond to genuine client requests. The interruption has an impact on the availability and performance of the targeted service or application, causing inconvenience or financial loss.

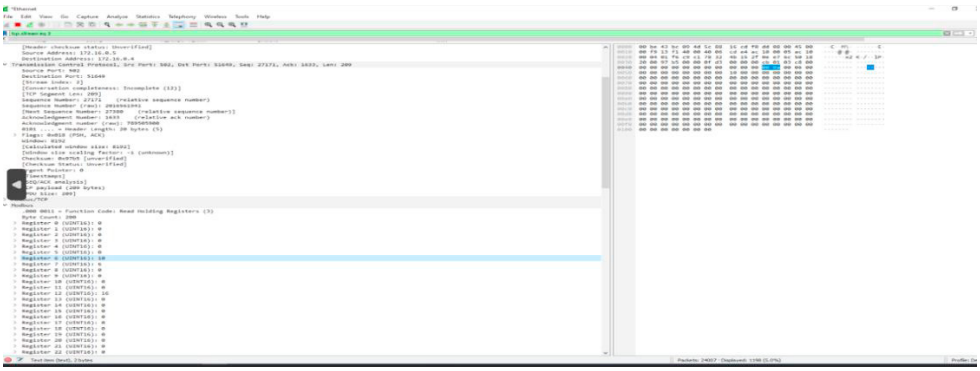

Fig. 9. Data through Modbus without encryption

6. Conclusion

Modbus-based ICS systems are more prone to malicious attacks because of a lack of vulnerabilities. Our paper emphasized the significance of protecting data transfer within the Modbus protocol and highlighted the existing vulnerabilities in Modbus-based Industrial Control Systems (ICS). We discovered the lack of encryption in the Modbus protocol as a serious weakness, rendering it vulnerable to numerous forms of attacks, using real-time vulnerability assessment. Our findings highlight the critical need to improve the security mechanisms of Modbus-based ICS systems. Future research should concentrate on the development of specialized lightweight cryptographic algorithms and the advancement of security mechanisms Considering the resource limits of ICS components to preserve the integrity and confidentiality of data in Modbus-based ICS systems, guaranteeing their continuous functioning in critical industrial environments.

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